

Technical Dossier

FishyFinds

Internet Software Architectures (ISA) · 2021/22

Dossier 1.0.0 · verified 9 September 2026 · smoke 66/66

Full technical write-up of the fishing-tourism marketplace: purpose of the subject and project, stack, architecture, domain rules, role UI with annotated screenshots, concurrency, tests, scalability, and summary.

Spring Boot 2.5.7 + Vue 2 + PostgreSQL

Local: <http://localhost:8080>

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Document control

Field	Value
Title	Technical Dossier, FishyFinds (ISA)
Dossier version	1.0.0
Software	FishyFinds / artifact com.fishyfinds:isa 0.0.1-SNAPSHOT
Course	Internet Software Architectures (ISA), FTN UNS, 2021/22
Compiled	9 September 2026
Authors	Natalija Simin (Guest / Customer), Laslo Uri (Bungalow / Boat owner), David Jandrić (Instructor / Admin)
Institution	Faculty of Technical Sciences, University of Novi Sad, Computing and Control Engineering
Live check	scripts/full-smoke.ps1 66/66 PASS ; Maven service tests green (JDK 17)
Local entry	http://localhost:8080

1. Purpose of the subject

Internet Software Architectures (ISA) is the FTN course where a three-student team designs and ships a multi-role web system with a real persistence layer, authentication, and graded non-functional topics (concurrency, DevOps, scalability). The exam expects:

- a shared product with **role-split ownership** (each student defends a persona),
- REST + SPA (or equivalent) over a relational database,
- documented concurrent-access behaviour,
- automated tests and CI,
- a short scalability argument with at least one concrete mechanism in code.

This dossier is the archive write-up for that defense: what the system is for, how it is built, what each grade band covers, and what the live UI looked like when verified in September 2026.

2. Purpose of the project

FishyFinds is a fishing-tourism marketplace. Guests browse coastal bungalows, boats, and instructor-led courses. Registered customers reserve and cancel slots. Advertisers (bungalow owners, boat owners, instructors) publish offers, manage availability, run discount “quick actions,” book for walk-in clients, and read occupancy / income charts. Administrators moderate registrations, reviews, complaints, account deletions, no-show penalties, the loyalty ladder, the user/offer directory, and the platform income cut.

The product is intentionally one Spring Boot process serving a Vue 2 SPA from static/, backed by PostgreSQL. It is a course system, not a hosted production SaaS: SMTP may be unset, passwords are demo-level, and schema is recreated from seed on boot (create-drop + data-postgres.sql).

3. Identification

Item	Detail
Display name	FISHYFINDS (red fish mark, black wordmark)
English product name	FishyFinds — coastal stays & fishing days
Serbian product name	FishyFinds — obalski smeštaj i ribolovni dani
Kind	Multi-role web application (SPA + REST)
Backend	Java 11 target, Spring Boot 2.5.7 , runs on JDK 17+
Frontend	Vue 2 (CDN), Vue Router history mode, Axios, SweetAlert2
Maps / charts	OpenLayers 6, Chart.js
Persistence	Spring Data JPA, Hibernate, PostgreSQL database fishyfinds_db
Auth	Spring Security + JWT (TokenUtils), roles via @PreAuthorize
Mail	Spring Mail (verification, booking, admin decisions; failures non-fatal)
Build	Maven Wrapper; GitHub Actions; SonarCloud
Tests	JUnit 5 + Mockito; PowerShell live smoke (scripts/full-smoke.ps1)
UI language	English chrome (course SPA)
Doc languages	English (this file) · Serbian Latin (DOSSIER_SR.md)

Author	Course roles	Main surfaces
Natalija Simin	Guest, Customer	Catalogs, register/activate, search & book, cancel (≥3 days), history, feedback, follow, complaints, penalties, profile
Laslo Uri	Bungalow owner, Boat owner	Offer CRUD, terms, quick actions, book-for-client, maps, unavailability calendar, /owner-reports
David Jandrić	Instructor, Admin	Courses, visit reports, loyalty, admin queues (registrations, reviews, complaints, deletions, penalties, directory, income)

Seed password for all demo accounts: **password**.

Email	Role
mail@mail.com	Customer
zokaMagic@mail.com	Bungalow owner
zokiSumi@mail.com	Boat owner
vesnaVuki@mail.com	Instructor
admin@admin.com	Admin
pending.owner@mail.com	Pending registration

4. Scope

In scope: browsing three offer types; JWT login; customer booking/cancel; owner/instructor offer lifecycle (create, edit blocked while reserved, soft-delete, terms, quick actions, book-for-client); OpenLayers maps; occupancy / unavailability calendar; Chart.js reports; loyalty configuration; full admin moderation hub; concurrent-access write-ups; service tests; catalog caching for the scalability PoC.

Out of scope / honest limits: production SMTP hardening, multi-tenant hosting, pixel-perfect capture of every nested admin modal, and a full human click-through of every SweetAlert path. Live smoke covers **66** API/UI-critical checks across roles; that is the verification bar used for this dossier.

5. Technology stack (explained)

Layer	Choice	Why it is here
Spring Boot 2.5.7	Application container	Course-era stack: embedded Tomcat, auto-config for Web, Security, JPA, Mail
Spring Web / Data REST	HTTP surface	Controllers under controllers/ expose /api/**; SPA is static
Spring Security + JWT	AuthN/Z	Stateless API: login returns token; filter parses Authorization; method security by role
Spring Data JPA / Hibernate	ORM	Maps User hierarchy and Offer subtypes to PostgreSQL; @Version for optimistic locking
PostgreSQL	RDBMS	Relational integrity for reservations overlapping terms; seed via data-postgres.sql
Spring Mail	Notifications	Activation, booking, subscriber quick-action mail; wrapped so missing SMTP does not fail bookings
Vue 2 CDN SPA	UI	Fast course delivery without a Node build pipeline; components in static/
Vue Router (history)	Client routes	Pretty URLs; SpaForwardController forwards unknown GETs to index.html
Axios	HTTP client	Attaches JWT; talks to /api
OpenLayers 6	Maps (grade 7)	Offer detail maps from Location coordinates
Chart.js	Analytics (grade 7)	Owner occupancy / income charts; admin finance views
Spring Cache @Cacheable	Scalability PoC (grade 10)	Catalog findAll for bungalows / boats / courses
Maven + GHA + Sonar	DevOps (grade 9)	Build, test, static analysis

Source layout

```
src/main/java/com/fishyfinds/isa/  
  controllers/    REST (offers, users, terms, admin, analytics...)  
  service/       Business rules (reservation, mail, loyalty...)  
  model/         JPA entities + enums  
  repository/    Spring Data interfaces  
  security/      JWT filter, TokenUtils  
  config/        Security, cache, SPA forward  
src/main/resources/static/      Vue SPA (kebab-case pages)  
src/main/resources/data-postgres.sql  
docs/concurrent-access/        Grade 8 PDFs + MD  
docs/scalability/SCALABILITY_POC.md Grade 10 write-up  
scripts/full-smoke.ps1         Live verification
```

6. System architecture

Architecture

Figure 1. Browser Vue SPA → Spring Boot (:8080) → controllers/services → JPA → PostgreSQL. JWT travels in localStorage and the Authorization header. Catalog reads may hit Spring Cache.

Use cases

Figure 2. Role split matching the three ISA students, plus shared grade 7–10 concerns.

Reservation flow

Figure 3. Customer (or owner-for-client) booking path: controller → service checks → optimistic/pessimistic locks → commit → best-effort mail.

Runtime is a **single JVM**. There is no separate Node server. Static assets and REST share origin `localhost:8080`, which simplifies CORS for the course demo.

7. Domain model (application detail)

Users. User implements `UserDetails`. Subtypes: `Customer`, `BungalowOwner`, `BoatOwner`, `Instructor`, `Admin`, plus `Authority` / registration status. Owners and instructors register as advertisers and wait for admin approval; customers activate via mail token when SMTP is configured.

Offers. Abstract `Offer` with `Location`, `ImageItem`, `AdditionalService`, `Price`. Concrete types: `Bungalow`, `Boat` (with `Engine`), `Course` (instructor link). Soft-delete and “no edit while active reservations” protect booked inventory.

Scheduling. Term is a bookable window on an offer (`@Version`). `Reservation` binds a customer (or null for `ReservationType.QUICK`) to a term. `CancelledReservation` records cancels. `OfferUnavailability` blocks calendar ranges so bookings cannot overlap closed periods (grade 7).

Moderation & loyalty. `Complaint`, `UserFeedback`, `AccountDeletionRequest`, `Penal`, `LoyaltyProgram`, per-role loyalty rows (`CustomerLoyalty`, `OwnerLoyalty`, `InstructorLoyalty`), `VisitReport`, `Subscriber` (follow offer → mail on quick action).

Key rules coded in services

- Cancel only if the reservation starts in ≥ 3 days.
- `createQuickAction` creates a quick reservation and emails subscribers (`MailService.sendNewActionEmail`).
- `updateBungalow|Boat|Course/{id}` refused when active reservations exist.
- Terms returned by `getTermsByOfferId` only if `endTime` is still after “now” (seed windows pushed to **2027** so the 2026 archive run stays bookable).
- Missing customer `Penal` row is auto-created on booking so penalty flows never NPE.
- Duplicate `DELETE /api/deleteUser/{id}` mapping was removed; directory delete stays on `AdminDirectoryController`.

8. Functional specification by role

Guest. Landing hero; browse bungalows / boats / courses; open details (description, price, rating, map); register; sign in.

Customer. Search/sort catalogs; reserve a free term; see upcoming / history; cancel with the 3-day rule; leave feedback; follow offers for actions; file complaints; view penalties; edit profile.

Bungalow / boat owner. My listings with Details / Terms / Edit / Actions / Book for client / Delete; add terms via daterange UI; publish quick actions; mark unavailability on calendar; open /owner-reports charts; visit report entry points where applicable.

Instructor. Same offer-management parity on courses (/my-courses): terms, actions, book-for-client, calendar, reports.

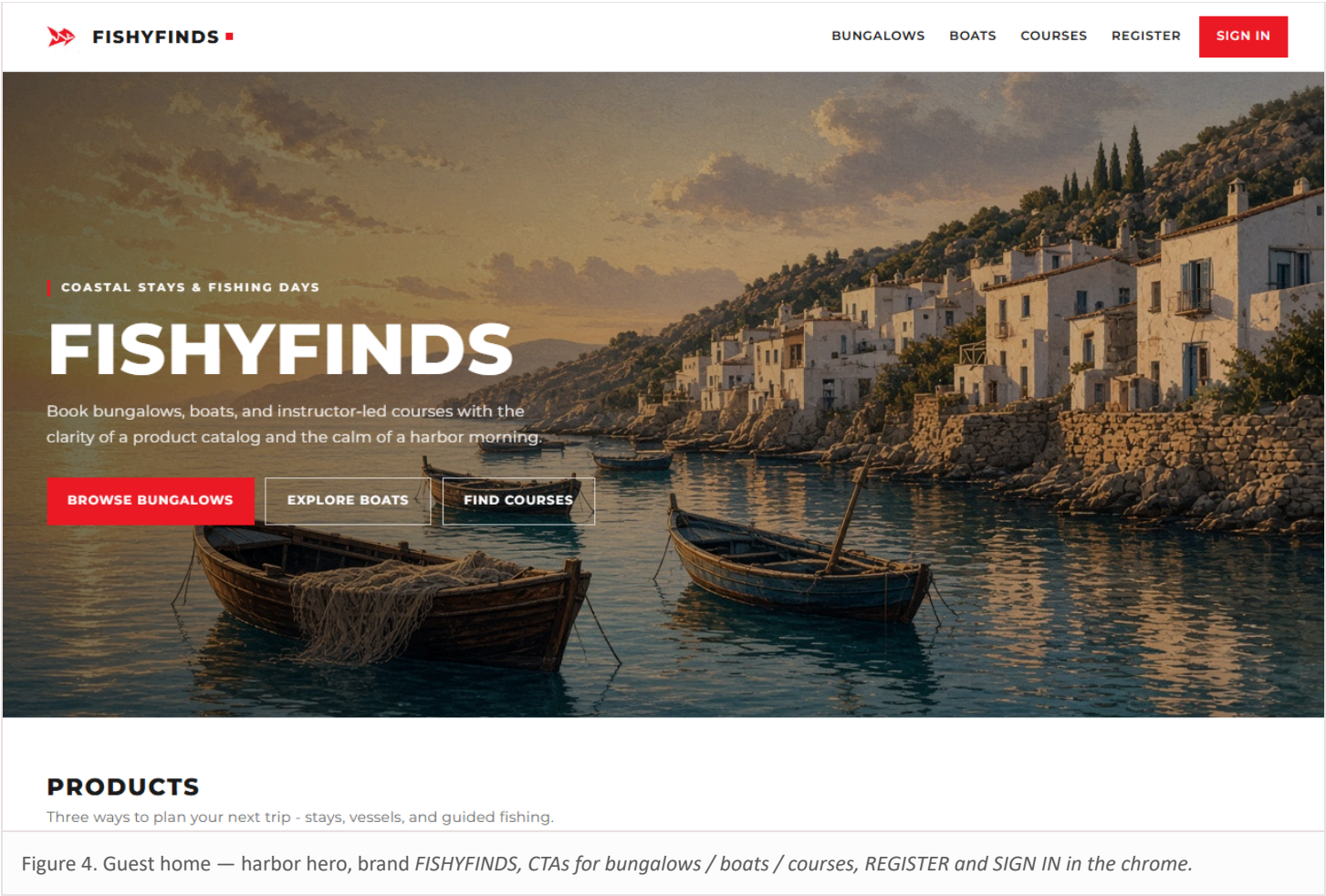
Admin. Dashboard tiles: registrations, complaints, reviews, deletion requests, penalties, directory, register admin, loyalty, income / system cut. Each tile opens a moderation queue with accept/deny style decisions and optional mail.

9. Grade ladder (ISA 6 → 10)

Grade	Expectation	What is in this tree
6	Core §3 marketplace	Terms UI for all offer types; POST /api/createQuickAction + subscriber mail; PUT update endpoints; book-for-client; soft-delete
7	Maps, charts, calendar, loyalty	OpenLayers on details; Chart.js reports; OfferUnavailability; /admin-loyalty
8	Concurrent access	PDFs under docs/concurrent-access/ (Students 1–3); @Version + locks in services
9	DevOps + tests	GitHub Actions, SonarCloud; Mockito tests for reservation create/cancel, quick action, loyalty CRUD, penal approve
10	Scalability	docs/scalability/SCALABILITY_POC.md; @Cacheable on catalog findAll

10. User interface (live captures)

Screens captured from the local app on 9 September 2026 against seeded data. Images live in [../screenshots/](#).



BUNGALOWS

Browse coastal stays - search, sort, and open details.

NAME	LOCATION	AVAILABLE FROM	AVAILABLE UNTIL
e.g. Coastal Cabin	City or country	mm/dd/yyyy --:--	mm/dd/yyyy --:--

SORT RESULTS

Choose order

SEARCH

CLEAR

Leave dates empty to browse all bungalows. Sort updates the list immediately.



NIKOS
25 | * 3.5
Nikos: seaside/mountain bungalow with 1 rooms and 2 beds. Walk to water or trail.

DETAILS

FOLLOW



SHREK HOUSE
32.5 | * 3.6
Shrek house: seaside/mountain bungalow with 2 rooms and 3 beds. Walk to water or trail.

DETAILS

FOLLOW



FIONA
40 | * 3.7
Fiona: seaside/mountain bungalow with 3 rooms and 4 beds. Walk to water or trail.

DETAILS

FOLLOW



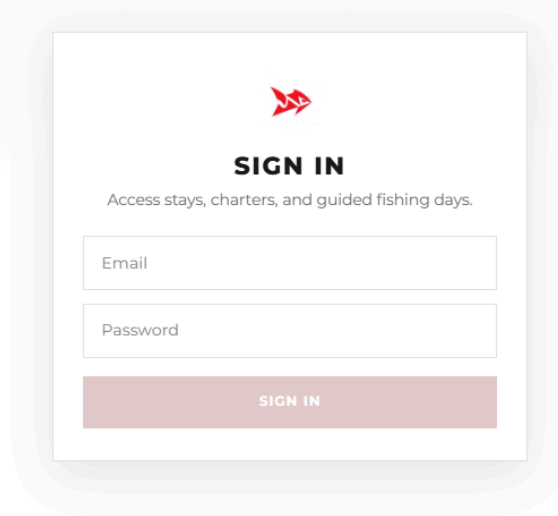
AZURE CABIN
47.5 | * 3.8
Azure Cabin: seaside/mountain bungalow with 1 rooms and 5 beds. Walk to water or trail.

DETAILS

FOLLOW



Figure 5. Public bungalow catalog — card grid with photo, price, rating, location; entry point for guest browsing and customer search.



The sign-in form is centered on the page. It features the FishyFinds logo at the top, followed by the heading 'SIGN IN' and a subtext 'Access stays, charters, and guided fishing days.' Below this are two input fields: 'Email' and 'Password'. At the bottom of the form is a red 'SIGN IN' button.



SIGN IN

Access stays, charters, and guided fishing days.

SIGN IN

Figure 6. Sign-in — JWT login against `/api` authentication; demo accounts use password `password`.

BOATS

Charters and day boats - search, sort, and reserve.

NAME	LOCATION	AVAILABLE FROM	AVAILABLE UNTIL
e.g. Sea Breeze	City or country	mm/dd/yyyy --:--	mm/dd/yyyy --:--
SORT RESULTS			
Choose order		SEARCH	CLEAR
Leave dates empty to browse all boats. Sort updates the list immediately.			



TITANIK

40 | * 3.8

Titanik: fishing skiff (5.5m) for coastal fishing and leisure. Engine #1.

DETAILS

FOLLOW



BLUE MARLIN

52.5 | * 3.9

Blue Marlin: cabin cruiser (6.2m) for coastal fishing and leisure. Engine #2.

DETAILS

FOLLOW



SEA BREEZE

65 | * 4

Sea Breeze: sailboat (6.9m) for coastal fishing and leisure. Engine #3.

DETAILS

FOLLOW



CAPTAIN'S PRIDE

77.5 | * 4.1

Captain's Pride: pontoon (7.6m) for coastal fishing and leisure. Engine #4.

DETAILS

FOLLOW

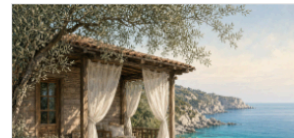
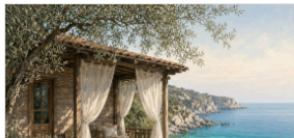


Figure 7. Boats catalog — same product-card pattern as bungalows; owner role later manages these via My boats.

COURSES

Instructor-led fishing adventures - browse and book.

COURSE NAME	LOCATION	INSTRUCTOR	AVAILABLE FROM	AVAILABLE UNTIL
e.g. Fly fishing intro	City or country	Instructor name	mm/dd/yyyy --:--	mm/dd/yyyy --:--

SORT RESULTS

Leave dates empty to browse all courses. Sort updates the list immediately.



RACING WITH THE HIPPOS

Vesna Vukelic | 18 | * 4

Racing with the hippos: instructor-led fishing adventure suitable for 2 guests. Gear briefing included.

[DETAILS](#)[FOLLOW](#)

DAWN SHORE CASTING

Zeljana Cudesnic | 24 | * 4.1

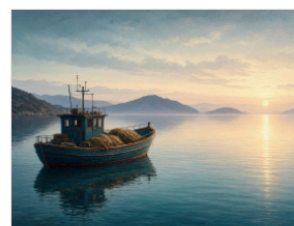
Dawn Shore Casting: instructor-led fishing adventure suitable for 3 guests. Gear briefing included.

[DETAILS](#)[FOLLOW](#)

KIDS FIRST CATCH

Nemanja Udica | 30 | * 4.2

Kids First Catch: instructor-led fishing adventure suitable for 4 guests. Gear briefing included.

[DETAILS](#)[FOLLOW](#)

SPINNING MASTERCLASS

Kristina Mreza | 36 | * 4.3

Spinning Masterclass: instructor-led fishing adventure suitable for 5 guests. Gear briefing included.

[DETAILS](#)[FOLLOW](#)

Figure 8. Courses catalog — instructor-led fishing courses exposed to guests and customers.

MAKE A RESERVATION

Search available terms by type, date, duration, and party size.

OFFER TYPE	START DATE	DURATION (DAYS)	GUESTS	
Bungalow	mm/dd/yyyy --:-- 	e.g. 3	e.g. 2	SEARCH
Fill all fields to find free terms that match your stay.				

No available terms yet - adjust filters and search.

Figure 9. Reservation flow UI — customer selects a free term on an offer and confirms booking (service enforces availability and unavailability rules).

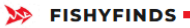
UPCOMING RESERVATIONS

Your future bookings - search, sort, filter, or cancel.

OFFER NAME	SORT RESULTS	TYPE	
Search bookings	Choose order	All types	SEARCH
Sort and type filter apply immediately.			

 WAVE DANCER 2027-09-09T10:00:00 - 2027-09-11T10:00:00 Wave Dancer: fishing skiff (9.7m) for coastal fishing and leisure. Engine #7. People: 1 Total: 390 * 4.4 CANCEL RESERVATION	 CYPRESS REST 4 2026-10-05T10:00:00 - 2026-10-08T10:00:00 Cypress Rest 4: seaside/mountain bungalow with 2 rooms and 2 beds. Walk to water or trail. People: 2 Total: 520 * 3.9 CANCEL RESERVATION	 SEA BREEZE 3 2027-01-09T10:00:00 - 2027-01-12T10:00:00 Sea Breeze 3: sailboat (11.1m) for coastal fishing and leisure. Engine #39. People: 2 Total: 840 * 4 CANCEL RESERVATION	 ADVANCED JIGGING 2027-04-15T10:00:00 - 2027-04-18T10:00:00 Advanced Jigging: instructor-led fishing adventure suitable for 4 guests. Gear briefing included. People: 2 Total: 1160 * 4.2 CANCEL RESERVATION
--	--	--	--

Figure 10. Customer upcoming reservations — cancel is offered only when the ≥ 3 -day rule still holds.



MY BUNGALOWS CALENDAR REPORTS VISIT REPORT MY ACCOUNT SIGN OUT

MY BUNGALOWS

Search, sort, and manage your coastal bungalow listings.

BUNGALOW NAME

Search by name

LOCATION

City or country

SORT RESULTS

Choose order

SEARCH

ADD BUNGALOW



NIKOS

25 \$ | * 3.5 | Leptokaria

Nikos: seaside/mountain bungalow with 1 rooms and 2 beds. Walk to water or trail.

DETAILS

TERMS

EDIT

ACTIONS

BOOK FOR CLIENT

DELETE



PINE HAVEN 2

40 \$ | * 4.2 | Rovinj 2

Pine Haven 2: seaside/mountain bungalow with 2 rooms and 4 beds. Walk to water or trail.

DETAILS

TERMS

EDIT

ACTIONS

BOOK FOR CLIENT

DELETE



SEAGLASS COTTAGE 3

55 \$ | * 4.9 | Igoumenitsa 4

Seaglass Cottage 3: seaside/mountain bungalow with 3 rooms and 2 beds. Walk to water or...

DETAILS

TERMS

EDIT

ACTIONS

BOOK FOR CLIENT

DELETE



SUNSET LOFT 4

70 \$ | * 4.1 | Dubrovnik

Sunset Loft 4: seaside/mountain bungalow with 1 rooms and 4 beds. Walk to water or trail.

DETAILS

TERMS

EDIT

ACTIONS

BOOK FOR CLIENT

DELETE



Figure 11. My bungalows (Laslo / bungalow owner) — search/sort, ADD BUNGALOW, and per-card Details / Terms / Edit / Actions / Book for client / Delete.

MY BOATS

Search, sort, and manage your boat listings.

BOAT NAME	LOCATION	SORT RESULTS		
Search by name	City or country	Choose order	SEARCH	ADD BOAT



TITANIK
40 \$ | * 3.8 | Zlatibor
Type: fishing skiff | Length: 5.5 m | Engines: 2 | Power: 100 | Max speed: 35
Titanik: fishing skiff (5.5m) for coastal fishing and leisure. Engine #1.



SILVER HOOK 2
115 \$ | * 4.8 | Volos 2
Type: speedboat | Length: 6.9 m | Engines: 3 | Power: 540 | Max speed: 145
Silver Hook 2: speedboat (6.9m) for coastal fishing and leisure. Engine #23.



PELICAN SCOUT 3
90 \$ | * 4.6 | Kragujevac 4
Type: sailboat | Length: 8.3 m | Engines: 1 | Power: 980 | Max speed: 255
Pelican Scout 3: sailboat (8.3m) for coastal fishing and leisure. Engine #45.



LUCKY NET 4
65 \$ | * 4.4 | Pula
Type: fishing skiff | Length: 9.7 m | Engines: 2 | Power: 1420 | Max speed: 365
Lucky Net 4: fishing skiff (9.7m) for coastal fishing and leisure. Engine #67.

Figure 12. My boats — boat-owner parity with the bungalow management surface.

MY COURSES

Search, sort, and manage the courses you teach.

COURSE NAME	LOCATION	SORT RESULTS		
Search by name	City or country	Choose order	SEARCH	ADD COURSE



RACING WITH THE HIPPOS
18 \$ | * 4 | Cap 2
Ohrid, North Macedonia
Racing with the hippos: instructor-led fishing adventure suitable for 2 guests. Gear briefing included.



FLY FISHING INTRO 2
42 \$ | * 4.2 | Cap 4
Cavtat 2, Croatia
Fly Fishing Intro 2: Instructor-led fishing adventure suitable for 4 guests. Gear briefing included.



REEF BOTTOM FISHING 3
66 \$ | * 4.4 | Cap 6
Zadar 5, Croatia
Reef Bottom Fishing 3: instructor-led fishing adventure suitable for 6 guests. Gear briefing included.



ADVANCED JIGGING 4
36 \$ | * 4.6 | Cap 3
Novi Sad, Serbia
Advanced Jigging 4: instructor-led fishing adventure suitable for 3 guests. Gear briefing included.



Figure 13. My courses (instructor) — course CRUD and the same term/action/book tools as other advertisers.

OCCUPANCY CALENDAR

Week strip, month grid, or yearly overview of your bookings and unavailable periods.

MARK UNAVAILABLE

OFFER

Moonlit Lodge 5

FROM

mm/dd/yyyy --:--



UNTIL

mm/dd/yyyy --:--



SAVE UNAVAILABLE PERIOD

WEEK

MONTH

YEAR

PREV

TODAY

NEXT

September 2026 · Bookings vs unavailable blocks

MON	TUE	WED	THU	FRI	SAT	SUN
31	1	2	3	4	5	6

Figure 14. Owner calendar — occupancy plus OfferUnavailability blocks so closed ranges cannot be booked.

OWNER REPORTS

Rating, income, and reservation volume at a glance.

AVERAGE RATING 4.3	INCOME 4197 \$
WEEKLY RESERVATIONS 0	MONTHLY RESERVATIONS 0
YEARLY RESERVATIONS 0	



Figure 15. Owner reports — Chart.js business charts (occupancy / income style analytics for the advertiser).

 **FISHYFINDS**

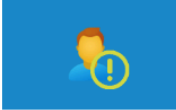
MODERATIONFINANCEMY ACCOUNTSIGN OUT

ADMIN DASHBOARD

Manage registrations, complaints, reviews, penalties, and platform settings.

**REGISTRATIONS**

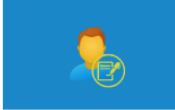
Approve or deny pending advertiser registrations

**COMPLAINTS**

Review and resolve customer complaints

**REVIEWS**

Accept or decline pending feedback

**DELETION REQUESTS**

Handle account deletion requests

**PENALTIES**

Approve or decline no-show penalty reports

**DIRECTORY**

Browse and manage users and offers

**REGISTER ADMIN**

Create another administrator account

**LOYALTY**

Configure loyalty categories and rates

**INCOME**

System cut and income overview

Figure 16. Admin dashboard — tile hub for registrations, complaints, reviews, deletions, penalties, directory, register-admin, loyalty, income.

ADMIN DASHBOARD

Manage registrations, complaints, reviews, penalties, and platform settings.

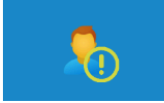
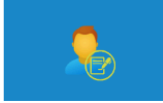
 REGISTRATIONS Approve or deny pending advertiser registrations	 COMPLAINTS Review and resolve customer complaints	 REVIEWS Accept or decline pending feedback	 DELETION REQUESTS Handle account deletion requests	 PENALTIES Approve or decline no-show penalty reports	 DIRECTORY Browse and manage users and offers
 REGISTER ADMIN Create another administrator account	 LOYALTY Configure loyalty categories and rates	 INCOME System cut and income overview			

Figure 17. Registrations queue — approve or deny pending advertiser registrations.

Figure 18. Complaints queue — resolve customer complaints with admin decision mail when SMTP is available.

SYSTEM INCOME

Platform cut and aggregated reservation income.

CURRENT SYSTEM CUT

5%

NEW SYSTEM CUT (%)

e.g. 10

UPDATE CUT

INCOME REPORT

TOTAL INCOME

0 \$

SYSTEM INCOME

0 \$

OWNER INCOME

0 \$

RESERVATIONS

0

Figure 19. Income / system cut — platform finance overview for the administrator.

LOYALTY CATEGORIES

Define point thresholds, discounts, and earning rates.

Category name	Required points	Category discount (%)	Earning rate
ADD CATEGORY			

UNBORN SHARK Required points: 0 Discount: 0% Earning rate: 0 DELETE CATEGORY	BABY SHARK Required points: 50 Discount: 20% Earning rate: 0 DELETE CATEGORY	PARENT SHARK Required points: 100 Discount: 40% Earning rate: 5 DELETE CATEGORY	GRANDPARENT SHARK Required points: 500 Discount: 60% Earning rate: 10 DELETE CATEGORY
--	--	---	---

Figure 20. Loyalty program — configure categories and point rates (grade 7 epic owned with the instructor/admin student).

11. Security, mail, and data

- Passwords stored with Spring Security hashing; API authorization is JWT + role checks.
- Registration for advertisers stays *PENDING* until admin approval.
- Mail is **best-effort**: booking and quick-action success paths do not depend on a working SMTP server (important for offline demos).
- Schema strategy for demos: Hibernate create-drop reloads `data-postgres.sql` each boot. Term dates in seed use **2027** windows. Sequence `setval` calls keep `Loyalty_program` / `penal` inserts from colliding with identity columns.

12. Concurrency, tests, and scalability

Concurrency (grade 8). Optimistic `@Version` on `Term` and `Reservation`; pessimistic locking on customer lookup where double-booking races matter. Narrative + diagrams: `docs/concurrent-access/` (Student 1 client, Student 2 owners, Student 3 admin/instructor).

Tests (grade 9). Service-level Mockito coverage for reservation create/cancel, quick action, loyalty CRUD, and penal approval. `./mvnw test` on JDK 17 is green. Live regression: `scripts/full-smoke.ps1` (66 checks).

Scalability (grade 10). Written PoC discusses partitioning, read replicas, load balancing, and caching. In code, catalog `findAll` endpoints are `@Cacheable` so repeated guest traffic does not always hit PostgreSQL.

13. Build and run (2026 check)

```
# PostgreSQL: database fishyfinds_db (defaults user/password root/root;  
# override with DB_USERNAME / DB_PASSWORD – do not commit secrets)  
  
./mvnw spring-boot:run  
# open http://localhost:8080  
  
./mvnw test  
powershell -File scripts/full-smoke.ps1  
powershell -File scripts/capture-screenshots.ps1
```

JDK 17+ recommended even though `pom.xml` declares Java 11. App serves SPA and API on port **8080**.

14. Summary

FishyFinds is the ISA 2021/22 team archive: a Spring Boot + Vue 2 + PostgreSQL marketplace for bungalows, boats, and fishing courses, with a clear three-student role split, JWT security, admin moderation, maps/charts/calendar/loyalty, concurrent-access documentation, automated tests/CI, and a caching-backed scalability note. Verified locally on 9 September 2026 with 66/66 smoke checks and green Maven tests. This dossier pairs English and Serbian Latin editions for the oral defense and for later rebuilds of the archive.

15. Related documents

Document	Path
Serbian dossier	<u>DOSSIER_SR.md</u>
Interactive HTML	<u>dossier.html</u>
Concurrent access	<u>../concurrent-access/</u>
Scalability PoC	<u>../scalability/SCALABILITY_POC.md</u>
Screenshots	<u>../screenshots/</u>
Diagrams	<u>../assets/diagrams/</u>

FishyFinds ISA technical dossier — English edition.